LEVOTHYROXINE

Empty stomach, same time daily

Narrow therapeutic index

CHEMICAL NAME Sodium (2S)-2-amino-3-[4-(4-hydroxy-3,5-diiodophenoxy)-3,5-diiodophenyl]propanoate (L-thyroxine, T4)

CLASS Synthetic thyroid hormone (T4)

TRADE NAMES Eltroxin · Euthyrox · Synthroid · Levoxyl

THERAPEUTIC USES / INDICATIONS

- Hypothyroidism (primary, secondary, post-thyroidectomy, post-radioiodine).
- Congenital hypothyroidism (urgent treatment in newborns).
- TSH suppression in differentiated thyroid cancer.
- Myxoedema coma (IV).

PHARMACODYNAMICS

- Replaces **thyroxine** → ↑ basal metabolic rate, oxygen consumption, protein synthesis, cardiac output, sensitivity to catecholamines.
- Essential for growth and brain development in children.

ADMINISTRATION

- **PO** on an empty stomach 30–60 min before breakfast (or at bedtime ≥3 h after food), with water.
- Separate from calcium, iron, antacids, PPIs, soy, coffee by at least 4 h.
- Infants: crush tablet in small amount of water/breast milk — not with soy formula.
- Keep same brand where possible.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Usually none at correct dose.
- Over-replacement: palpitations, tremor, sweating, heat intolerance, weight loss, insomnia, diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Angina, arrhythmias (AF), MI in elderly/IHD when started too high.
- Osteoporosis with long-term over-replacement.
- Adrenal crisis if given before correcting adrenal insufficiency.

DOSAGES

- **Adult (healthy, <60 yr):** 1.6 mcg/kg/day (usually 50–100 mcg to start).
- **Elderly or cardiac disease:** start 12.5–25 mcg daily, increase by 25 mcg every 4–6 weeks.
- **Congenital hypothyroidism:** 10–15 mcg/kg/day.
- **Pregnancy:** increase dose by ~25–30% as soon as pregnancy confirmed.
- Adjust by TSH every 6–8 weeks after change.

CONTRAINDICATIONS & CAUTIONS

- Untreated thyrotoxicosis.
- Untreated adrenal insufficiency (give steroids first).
- Acute MI (relative).
- Not for weight loss.

KEY INTERACTIONS

- **Calcium, iron, antacids, sucralfate, PPIs, orlistat, cholestyramine:** ↓ absorption.
- **Warfarin:** ↑ anticoagulant effect.
- **Enzyme inducers (rifampicin, carbamazepine, phenytoin):** ↑ requirements.
- **Oestrogens:** ↑ requirements.

PHARMACOKINETICS

- **Onset:** 3–5 days; full effect 4–6 weeks. Absorption 40–80% (empty stomach).
- Converted peripherally to active T3 (deiodinases); highly protein-bound.
- Excreted in urine and faeces. **t½:** ~7 days (euthyroid).

NURSING MANAGEMENT

- Give on empty stomach at the same time daily.
- Check apical pulse; report HR >100, chest pain, palpitations.
- Monitor TSH (and free T4) 6–8 weeks after changes, then yearly.
- Elderly/cardiac: start low, go slow.
- Neonatal screening follow-up.

PATIENT EDUCATION

- Take every morning on an empty stomach; wait 30–60 min before breakfast or coffee.
- Lifelong treatment; do not stop.
- Report palpitations, chest pain, tremor or weight loss.
- Tell your doctor if you become pregnant (dose usually needs increasing).

KEY POINTS

- Synthetic T4 for hypothyroidism.
- Elderly/IHD: start 12.5–25 mcg.
- Empty stomach; separate from Ca, Fe, PPIs.
- TSH checks every 6–8 weeks; t½ 7 days.

SOURCES CHECKED **[630]** Levothyroxine 100 micrograms Tablets (UK SmPC) **[631]** Levothyroxine sodium, oral (US label)

Full references p. 432–457